01 / EXECUTIVE ASSESSMENT

COMMODITYLENS / RELEASE 1.0

Clarity, tested.

A validation report.

A disciplined assessment of the gold intelligence platform: what works locally, what the analytical evidence shows, and what must still be verified before production.

LOCAL VALIDATION PASSED

Production acceptance remains open.

BACKEND TESTS	BROWSER CHECKS	AUDIT FINDINGS	SAMPLE ROWS
37	9	0	13,144
Financial invariants, API and ingestion	Includes two accessibility scenarios	Reported frontend dependency vulnerabilities	Synthetic fixtures; not market data

ASSESSMENT

Ready for a controlled staging review.

The application, calculations and core user flows are supported by local verification. The evidence does not yet establish a production deployment, an accepted live-data feed, or an executable trading edge.

10 OCTOBER 2026

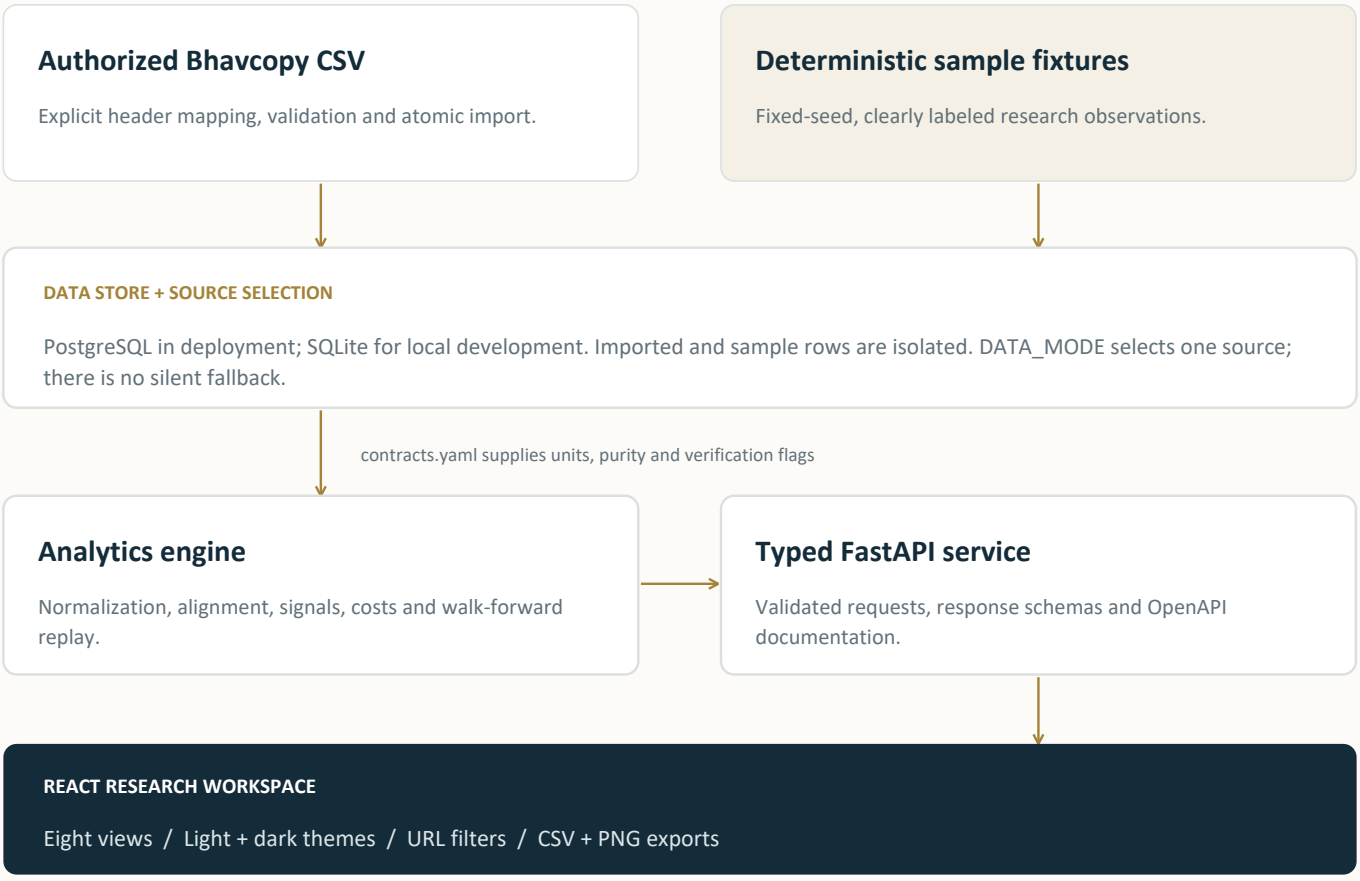
Prepared from repository evidence and reproducible sample analytics

Scope: internal engineering validation, not an independent audit or investment recommendation. The report contains exactly seven pages; sources and release gates appear on page 7.

02 / SYSTEM ARCHITECTURE

From source to decision.

Separate data origins. Explicit transformations. One traceable research workspace.



NINE-STEP RESEARCH FLOW

01 - 03

PREPARE

- Bhavcopy ingestion
- Cleaning & validation
- Contract identification

04 - 06

ANALYZE

- Fine-gold normalization
- Cross-contract comparison
- Signal generation

07 - 09

CHALLENGE

- Walk-forward backtest
- Cost & liquidity adjustment
- Trader dashboard

Architecture evidence: api/app, web/src and compose.yaml. The diagram describes the implemented design; the Docker/PostgreSQL runtime has not yet been executed on the development host.

03 / DATA INTEGRITY

The units must agree.

Normalization is auditable. Provenance and contract verification remain separate controls.

OBSERVATIONS

13,144

03 Mar 2025 - 09 Oct 2026

OBSERVED SESSIONS

420

Synthetic business-day fixtures

GOLD CONTRACTS

4

Configured outside the calculation logic

CONTRACT DEFAULTS - OPERATOR VERIFICATION PENDING

SYMBOL	LOT / G	QUOTE / G	PURITY	DELIVERY BASIS
GOLDM	100	10	995	Ahmedabad
GOLDTEN	10	10	999	Ahmedabad
GOLDGUINEA	8	8	999	Ahmedabad
GOLDPETAL	1	1	999	Mumbai

WORKED EXAMPLE / GOLDM / 09 OCT 2026 / SAMPLE DATA

INR 109,804.45

Settlement quote

→ Divide by 10 quotation grams

INR 10,980.44

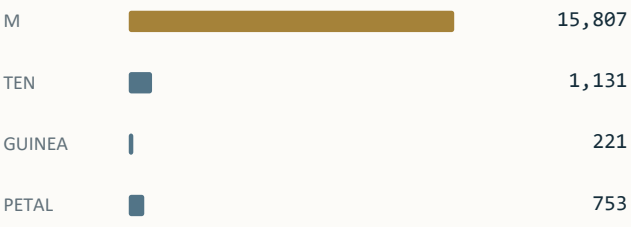
Per gram before purity

→ Then divide by 0.995 purity

INR 11,035.62

Per fine gram

LATEST SAMPLE PARTICIPATION



ENFORCED IMPORT CONTROLS

- Unambiguous dates; actual expiry retained
- Positive finite settlement; integral volume/OI
- Options excluded; duplicate conflicts rejected
- Atomic writes; SHA-256 repeat detection

Volume = traded lots. Zero-baseline bars; displayed prices rounded.

Source: sample database, contracts.yaml and ingestion tests. Defaults are illustrative until verified for each historical expiry. Gold Petal has a different delivery location; purity normalization does not remove that basis.

04 / SIGNAL VALIDATION

A deviation is not a trade.

Sample data | 09 October 2026 | October 2026 expiry | Prior 30-observation baseline

QUALIFIED SIGNALS

0

All current pairs remain below |z| = 2

WIDEST ABSOLUTE SPREAD

7.91 bp

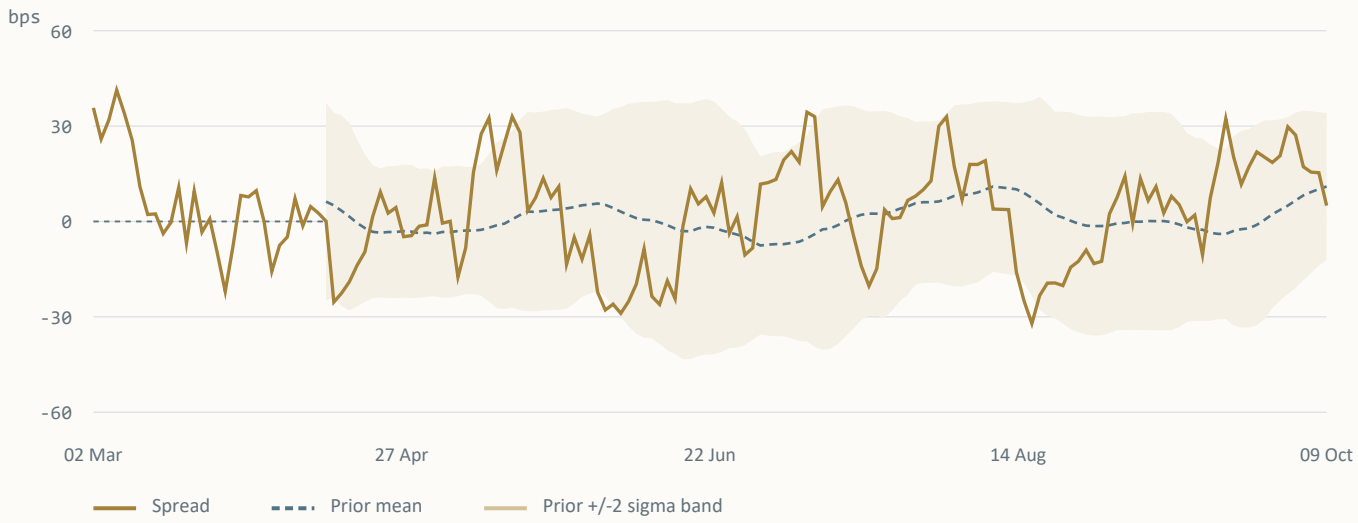
Across six canonical pairs

ANNUALIZED CARRY

6.48%

GOLDM: nearest two expiries

GOLDM / GOLDTEN NORMALIZED SPREAD



PAIRWISE Z-SCORES

	M	TEN	GUINEA	PETAL
M	0.00	-0.52	0.06	-0.39
TEN	0.52	0.00	0.38	0.01
GUINEA	-0.06	-0.38	0.00	-0.64
PETAL	0.39	-0.01	0.64	0.00

QUALIFICATION SEQUENCE

01 / Significance

Absolute z-score exceeds 2.

02 / Costs

Expected reversion exceeds round-trip cost.

03 / Liquidity

Each leg meets grade A or B.

04 / Comparability

Exact expiry and compatible delivery basis.

Chart points and matrix are computed from synthetic observations. Reverse cells are negative canonical-pair z-scores. Imported signals also require operator-verified specs. No qualified signal is fabricated when filters fail.

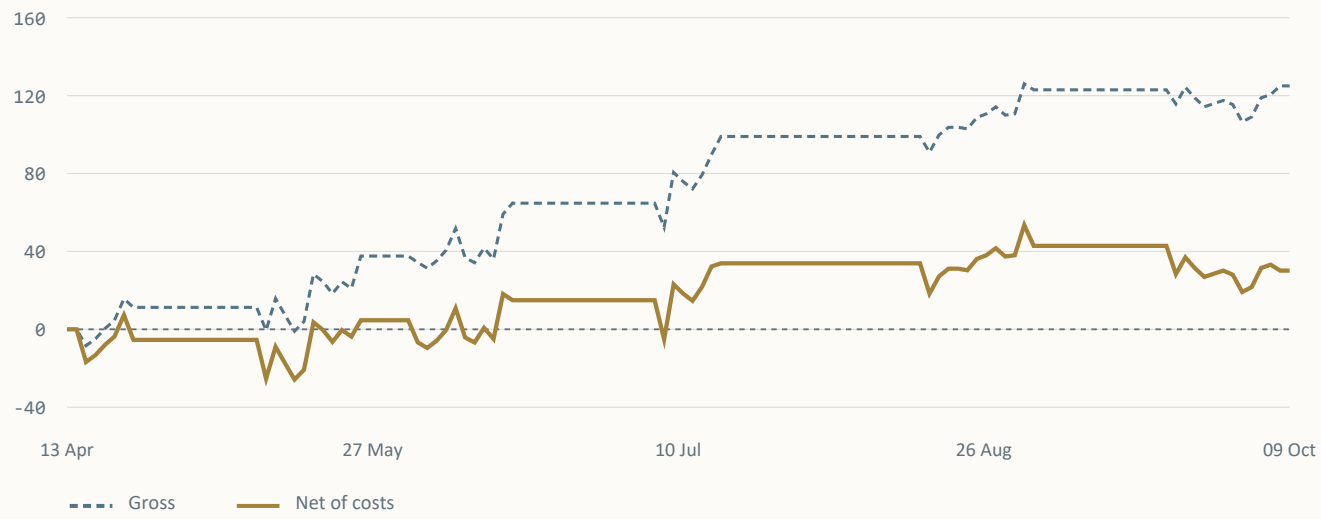
05 / BACKTEST EVIDENCE

Gross promise. Net reality.

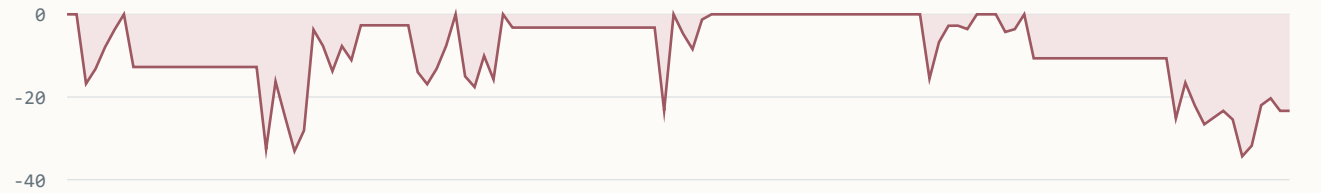
Default experiment | GOLDM / GOLDTEN | October 2026 expiry | Synthetic data only

GROSS P&L / BPS	NET P&L / BPS	MODELED COST / BPS
124.99	30.15	94.84
Before modeled costs	After incurred modeled costs	75.9% of gross P&L absorbed

CUMULATIVE RESEARCH P&L



NET DRAWDOWN / BPS



SHARPE	HIT RATE	CLOSED TRADES	MAX DRAWDOWN / BP
0.54	66.7%	6	-34.33

95% bootstrap interval: -59.12 to 139.03 bps

The interval crosses zero. Positive net simulation does not establish an executable edge.

Parameters: train 30 / test 10 sessions; entry |z| > 2; exit |z| <= 0.5; max hold 10; fees 1.5 bp/leg/side + slippage. 130 replay observations, 13 folds, 0 terminal positions. P&L basis: one-leg notional, fractional lots.

06 / ENGINEERING ASSURANCE

Test the claim, not the label.

Executed local checks support the stated behavior. Counts are not a code-coverage percentage.

BACKEND TESTS PASSED

37 / 37

Analytics, imports and API contracts

BROWSER CHECKS PASSED

9 / 9

Core flows, exports and failure states

ACCESSIBILITY VIEW/THEMES

10

Five key views in each of two themes

NO-LOOK-AHEAD: THE EXECUTION CLOCK

TRAIN

Prior 30 observations

→

TEST

Next 10 unseen sessions

→

SIGNAL

At the observed close

→

MODEL FILL

Next eligible open

Future prices are mutated and the dataset is truncated in tests. Earlier equity must remain identical. Missing opens and thin liquidity cannot create synthetic fills.

EVIDENCE REGISTER

Production build	TypeScript and optimized Vite build	PASSED
Accessibility automation	No detected A/AA violations on scoped views	PASSED
Dependency review	0 reported frontend production vulnerabilities	PASSED
Failure handling	API errors, empty source, invalid imports	PASSED
Container runtime	Docker / Nginx / PostgreSQL execution	NOT RUN

CHART PAYLOAD REDUCTION - COMPRESSED BUILD OUTPUT



Chart gzip payload reduced by 65.7%; the 503.50 kB chunk remains lazily loaded. Automated accessibility is not full WCAG certification. Tooling warnings: Plotly chunk size and Starlette test-client deprecation.

07 / RELEASE DECISION

Evidence before release.

The next milestone is staging acceptance, not a claim of zero defects or live trading readiness.

STAGING CANDIDATE

Seven release gates remain open.

01	**Container execution** Run Compose with PostgreSQL; exercise import/update and concurrent replay.	OPEN
02	**Hosting & access** Verify TLS, domain routing, trusted proxies, identity and monitoring.	OPEN
03	**Actual MCX files** Accept authorized files; verify symbol, expiry, option and OI semantics.	OPEN
04	**Historical specifications** Validate contract settings against every relevant expiry and delivery basis.	OPEN
05	**Execution assumptions** Calibrate fees/slippage; document lot, margin and market-depth limitations.	OPEN
06	**Recovery & capacity** Test encrypted backup/restore, rollback and latency/capacity budgets.	OPEN
07	**Accessibility review** Complete manual keyboard and screen-reader acceptance across all views.	OPEN

SOURCE AND REPRODUCIBILITY INDEX

S1	**Validation record**	docs/VERIFICATION.md; api/tests; web/tests
S2	**Analytical figures**	Recomputed from sample data on 10 Oct 2026; defaults, no tuning
S3	**Model boundaries**	docs/METHODOLOGY.md; docs/ACCEPTANCE.md
S4	**Official reference sources**	MCX product specifications and MCXCCL UDiFF schema; docs/SOURCES.md

Roadmap, not shipped services: mobile alerts, trained ML anomaly detection, billing, team roles, enterprise connectors and new markets. Independent audit, public hosting and actual-market validation are outside this report.